Algebra & Equations

Solving linear equations

1. Solve $3x - 7 = 11$.

2. Solve $5(x - 2) = 3x + 4$.

3. Solve $\frac{3x - 2}{4} = x - 1$.

4. Solve $2(x + 3) - 5 = 3(x - 1)$.

Expanding and simplifying expressions

5. Expand and simplify $3(2x - 1) - 2(x - 4)$.

6. Expand $(x + 5)(x - 2)$.

7. Expand $(2x - 3)^2$.

8. Simplify $\frac{6x^2 - 9x}{3x}$ (for $x \neq 0$).

Factoring quadratic expressions

9. Factor $x^2 + 7x + 12$.

10. Factor $x^2 - 2x - 15$.

11. Factor $2x^2 + 5x - 3$.

12. Factor $x^2 - 49$ using the difference of squares.

Solving quadratic equations by factoring

13. Solve $x^2 - 5x + 6 = 0$.

14. Solve $x^2 + 2x - 8 = 0$.

15. Solve $2x^2 - 5x - 3 = 0$.

16 Solve $x^2 - 9x + 20 = 0$.

The quadratic formula and the discriminant

17 Use the quadratic formula $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ to solve $x^2 + 3x + 1 = 0$, leaving the answer in exact form.

18 Find the discriminant of $2x^2 - 4x + 5 = 0$ and state how many real roots it has.

19 Find the discriminant of $x^2 - 6x + 9 = 0$ and state how many real roots it has.

20 Use the quadratic formula to solve $3x^2 - 2x - 1 = 0$.

Absolute value equations

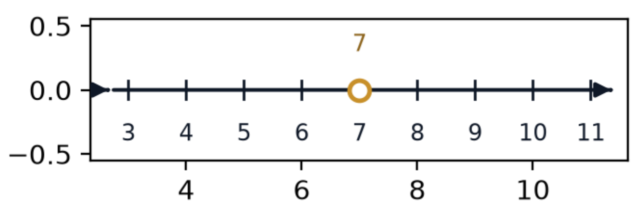
21 Solve $|x - 4| = 9$.

22 Solve $|2x + 3| = 7$.

23 Solve $3|x - 1| = 12$.

Linear inequalities

24 Solve $2x - 5 < 9$ and graph the solution set on a number line.



25 Solve $-3x + 4 \geq -8$.

26 Solve the compound inequality $5 < 2x + 1 \leq 11$.

Systems of linear equations

27 Solve the system $x + y = 10$, $x - y = 2$.

28 Solve the system $3x + 2y = 16$, $x - y = 2$.

29 This question has two parts. The sum of two numbers is 15 and their difference is 3.

a Write a system of two equations for the numbers x and y.

b Solve the system to find both numbers.

Rational equations

30 Solve $\dfrac{x}{x-2} = 3$.

31 Solve $\dfrac{2}{x+1} = \dfrac{1}{3}$.

32 Solve $\dfrac{x+1}{x-1} = 2$.

Solving for a variable in a formula

33 The area of a triangle is $A = \dfrac{1}{2}bh$. Solve for h.

34 The circumference of a circle is $C = 2\pi r$. Solve for r.

35 The formula $v = v_0 + at$ gives velocity under constant acceleration. Solve for t.

Linear-quadratic systems

36 Solve the system $y = x^2 - 1$, $y = 2x + 2$.

37 Solve the system $y = x^2$, $y = 3x - 2$.

Word problems: setting up and solving equations

38 This question has two parts. A rectangular garden has a length that is 4 meters more than its width w. Its area is 96 square meters.

a Write an equation relating w to the area.

b Solve for w, and state the garden's length.

39 This question has two parts. A mobile-data plan charges a fixed monthly fee f plus a rate r per gigabyte used. One month, 5 GB of data cost a total of 65 dollars; another month, 9 GB cost a total of 93 dollars.

a Write a system of two equations for f and r.

b Solve the system to find the fixed fee and the rate per gigabyte.

MCQ — Solving linear equations

40 Solve $4x - 9 = 15$.

a. $x = 6$

b. $x = -6$

c. $x = 24$

d. $x = 1.5$

41 Solve $6(x - 3) = 2x + 10$.

a. $x = -7$

b. $x = 4$

c. $x = 7$

d. $x = 1$

42 Solve $\dfrac{2x - 1}{3} = x - 2$.

a. $x = 5$

b. $x = 1$

c. $x = 7$

d. $x = -5$

43 Solve $3(x+4)-7=2(x-1)$.

a. $x=-7$

b. $x=-3$

c. $x=3$

d. $x=7$

44 Solve $5x+3=2x-9$.

a. $x=-2$

b. $x=2$

c. $x=4$

d. $x=-4$

MCQ — Expanding and simplifying expressions

45 Expand and simplify $4(3x-2)-3(x-5)$.

a. $15x-23$

b. $9x+23$

c. $9x+7$

d. $9x-23$

46 Expand $(x+7)(x-3)$.

a. $x^2+10x-21$

b. $x^2-4x-21$

c. $x^2+4x-21$

d. $x^2+4x+21$

47 Expand $(3x-1)^2$.

a. $3x^2-6x+1$

b. $9x^2-1$

c. $9x^2-6x+1$

d. $9x^2-6x-1$

48 Simplify $\frac{8x^2-12x}{4x}$ (for $x \neq 0$).

a. $2x-12$

b. $8x-3$

c. $2x-3$

d. $2x^2-3$

MCQ — Factoring quadratic expressions

49 Factor $x^2+9x+14$.

a. $(x+2)(x+7)$

b. $(x-2)(x-7)$

c. $(x+1)(x+14)$

d. $(x+14)(x-1)$

50 Factor $x^2-3x-18$.

a. $(x-9)(x+2)$

b. $(x-2)(x+9)$

c. $(x-6)(x+3)$

d. $(x+6)(x-3)$

51 Factor $3x^2 + 7x + 2$.

a. $(3x+1)(x+2)$

b. $(3x-1)(x-2)$

c. $(3x+2)(x+1)$

d. $(x+1)(3x+2)$

52 Factor $x^2 - 81$ using the difference of squares.

a. $(x-9)^2$

b. $(x-9)(x+9)$

c. $(x+9)^2$

d. $(x-81)(x+1)$

53 Factor $2x^2 - x - 6$.

a. $(2x+3)(x-2)$

b. $(2x-3)(x+2)$

c. $(x+3)(2x-2)$

d. $(2x+1)(x-6)$

MCQ — Solving quadratic equations by factoring

54 Solve $x^2 - 7x + 10 = 0$.

a. $x = 1$ or $x = 10$

b. $x = 7$ or $x = 10$

c. $x = -2$ or $x = -5$

d. $x = 2$ or $x = 5$

55 Solve $x^2 + 3x - 10 = 0$.

a. $x = -3$ or $x = 10$

b. $x = -10$ or $x = 1$

c. $x = 5$ or $x = -2$

d. $x = -5$ or $x = 2$

56 Solve $3x^2 - 11x - 4 = 0$.

a. $x = -\frac{1}{3}$ or $x = 4$

b. $x = \frac{1}{3}$ or $x = -4$

c. $x = 1$ or $x = 4$

d. $x = -\frac{1}{3}$ or $x = -4$

57 Solve $x^2 - 11x + 24 = 0$.

a. $x = 1$ or $x = 24$

b. $x = 4$ or $x = 6$

c. $x = -3$ or $x = -8$

d. $x = 3$ or $x = 8$

MCQ — The quadratic formula and the discriminant

58 Use the quadratic formula to solve $x^2 + 5x + 2 = 0$, leaving the answer in surd form.

a. $x = \frac{5 \pm \sqrt{17}}{2}$

b. $x = \frac{-5 \pm \sqrt{17}}{4}$

c. $x = \frac{-5 \pm \sqrt{33}}{2}$

d. $x = \frac{-5 \pm \sqrt{17}}{2}$

59 Find the discriminant of $3x^2 - 6x + 5 = 0$ and state how many real roots it has.

a. 96; two real roots

b. 24; two real roots

c. -24; no real roots

d. -24; one real root

60 Find the discriminant of $x^2 - 8x + 16 = 0$ and state how many real roots it has.

a. 0; one real root

b. 64; two real roots

c. 0; two real roots

d. -64; no real roots

61 Use the quadratic formula to solve $2x^2 + 3x - 2 = 0$.

a. $x = 1$ or $x = -2$

b. $x = -0.5$ or $x = 2$

c. $x = 0.5$ or $x = -2$

d. $x = 0.5$ or $x = 2$

MCQ — Absolute value equations

62 Solve $|x - 6| = 11$.

a. $x = -17$ or $x = 5$

b. $x = 5$ or $x = -17$

c. $x = 17$ or $x = -5$

d. $x = 17$ or $x = 5$

63 Solve $|3x+2| = 13$.

a. $x = \frac{11}{3}$ or $x = 5$

b. $x = 5$ or $x = -5$

c. $x = -\frac{11}{3}$ or $x = 5$

d. $x = \frac{11}{3}$ or $x = -5$

64 Solve $4|x-2| = 20$.

a. $x = 7$ or $x = -3$

b. $x = -7$ or $x = 3$

c. $x = 5$ or $x = -5$

d. $x = 7$ or $x = 3$

65 Solve $|2x-5| = 9$.

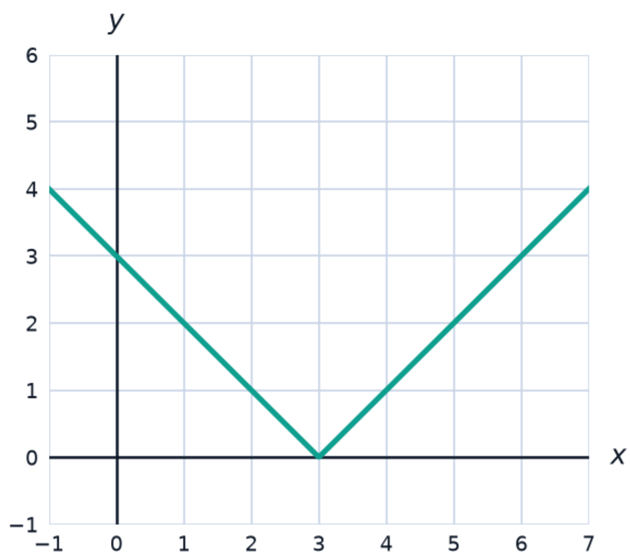
a. $x = 7$ or $x = 2$

b. $x = -7$ or $x = 2$

c. $x = 2$ or $x = -7$

d. $x = 7$ or $x = -2$

66 The graph shows $y = |x - 3|$. At what x-value does the graph reach its minimum?



a. $x = 1$

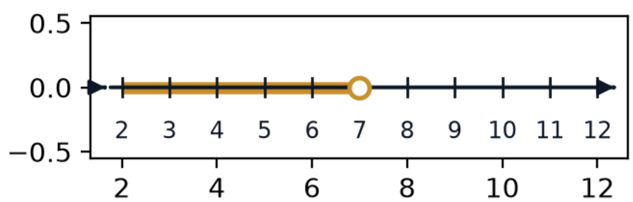
b. $x = 3$

c. $x = -3$

d. $x = 0$

MCQ — Linear inequalities

67 Solve $3x - 7 < 14$, and identify the graph of its solution set.



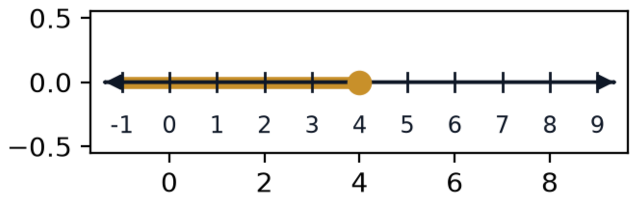
a. $x < 7$

b. $x \geq 7$

c. $x \leq 7$

d. $x > 7$

68 Solve $-4x + 6 \geq -10$, and identify the graph of its solution set.



a. $x > 4$

b. $x < 4$

c. $x \leq 4$

d. $x \geq 4$

69 Solve the compound inequality $7 < 3x + 1 \leq 19$.

a. $2 < x \leq 6$

b. $2 < x < 6$

c. $2 \leq x < 6$

d. $6 < x \leq 2$

70 Solve $2x + 5 > -3$ and $x - 1 < 6$ together (both must hold). Find the combined solution set.

a. $x > -4$

b. $-4 < x \leq 7$

c. $x < 7$

d. $-4 < x < 7$

MCQ — Systems of linear equations

71 Solve the system $x + y = 14$, $x - y = 4$.

a. $x = 5, y = 9$

b. $x = 10, y = 4$

c. $x = 9, y = 5$

d. $x = 9, y = 4$

72 Solve the system $4x + 3y = 26$, $x - y = 3$.

a. $x = 5, y = 2$

b. $x = 2, y = 5$

c. $x = 8, y = 5$

d. $x = 5, y = 3$

73 The sum of two numbers is 22 and their difference is 6. Find the two numbers.

a. 11 and 11

b. 14 and 6

c. 16 and 6

d. 14 and 8

74 Solve the system $2x - y = 7$, $3x + 2y = 21$.

a. $x = 5, y = -3$

b. $x = 5, y = 3$

c. $x = 3, y = 5$

d. $x = 7, y = 7$

MCQ — Rational equations

75 Solve $\dfrac{x}{x-3}=4$.

a. $x=12$

b. $x=-4$

c. $x=4$

d. $x=3$

76 Solve $\dfrac{3}{x+2}=\dfrac{1}{4}$.

a. $x=6$

b. $x=14$

c. $x=10$

d. $x=-10$

77 Solve $\dfrac{x+2}{x-2}=3$.

a. $x=2$

b. $x=8$

c. $x=4$

d. $x=-4$

78 Solve $\dfrac{4}{x-1}=\dfrac{2}{x+2}$.

a. $x=-5$

b. $x=5$

c. $x=-1$

d. $x=2$

79 Solve $\frac{6}{x} = \frac{2}{x-4}$.

a. $x = 12$

b. $x = 4$

c. $x = -6$

d. $x = 6$

MCQ — Solving for a variable in a formula

80 The perimeter of a rectangle is $P = 2l + 2w$. Solve for l.

a. $l = \frac{P}{2} - w$

b. $l = \frac{P - 2w}{2}$

c. $l = \frac{P + 2w}{2}$

d. $l = P - 2w$

81 The formula for simple interest is $I = Prt$. Solve for r.

a. $r = \frac{I}{P}$

b. $r = \frac{I}{Pt}$

c. $r = \frac{Pt}{I}$

d. $r = I - Pt$

82 The formula $F = \frac{9}{5}C + 32$ converts Celsius to Fahrenheit. Solve for C.

a. $C = \frac{5(F - 32)}{9}$

b. $C = \frac{5F - 32}{9}$

c. $C = \frac{5}{9}F + 32$

d. $C = \frac{9(F - 32)}{5}$

83 The volume of a cylinder is $V = \pi r^2 h$. Solve for h.

a. $h = \dfrac{V}{\pi r^2}$

b. $h = V - \pi r^2$

c. $h = \dfrac{V}{\pi r}$

d. $h = \dfrac{\pi r^2}{V}$

MCQ — Linear-quadratic systems

84 Solve the system $y = x^2 - 4$, $y = 3x$.

a. $(-4, -12)$ and $(1, 3)$

b. $(-1, -3)$ and $(1, 3)$

c. $(4, 12)$ and $(1, 3)$

d. $(4, 12)$ and $(-1, -3)$

85 Solve the system $y = x^2$, $y = 2x + 3$.

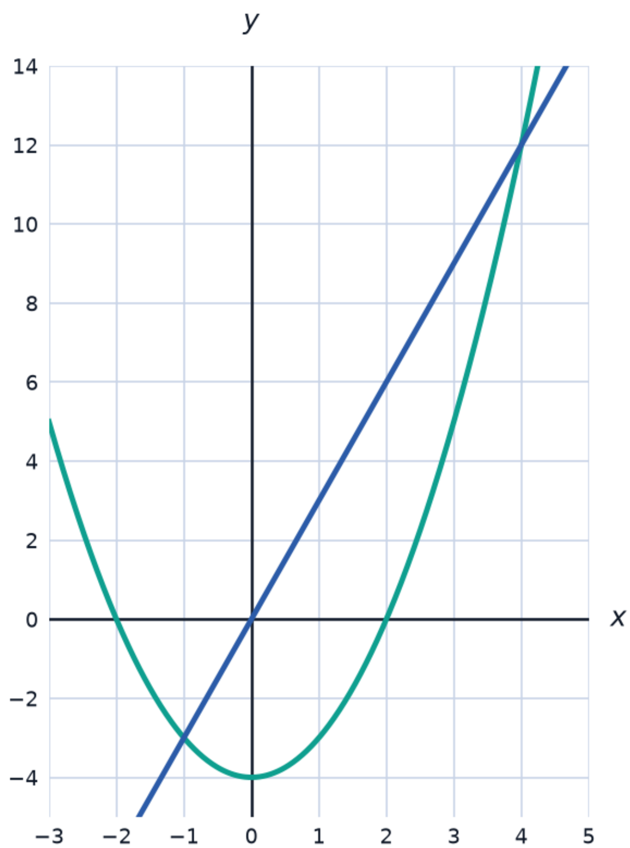
a. $(-3, 9)$ and $(1, 1)$

b. $(3, 9)$ and $(1, 1)$

c. $(-1, 1)$ and $(1, 1)$

d. $(3, 9)$ and $(-1, 1)$

86 The graph shows the system $y = x^2 - 4$, $y = 3x$. What are the points of intersection?



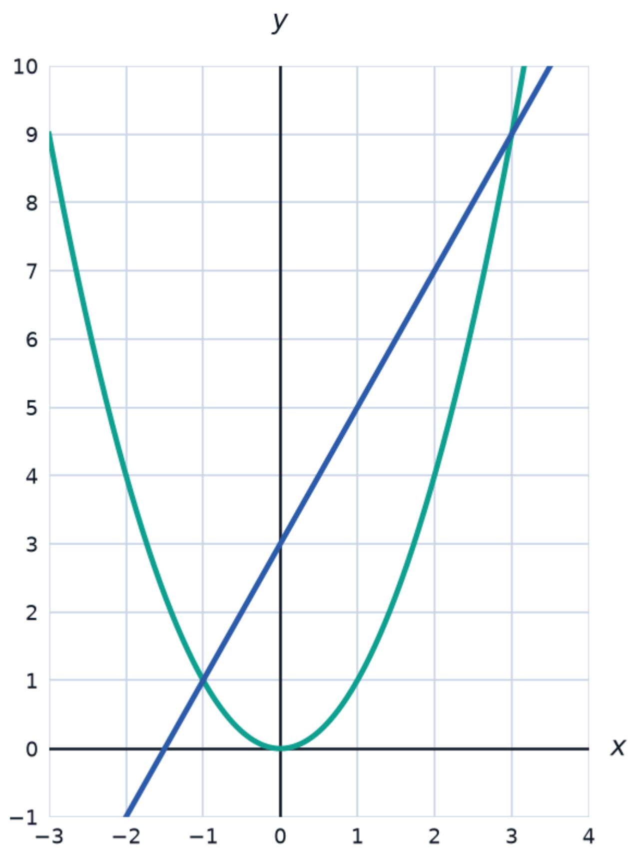
a. $(4, 12)$ and $(1, 3)$

b. $(-4, -12)$ and $(1, 3)$

c. $(-1, -3)$ and $(1, 3)$

d. $(4, 12)$ and $(-1, -3)$

87 The graph shows the system $y = x^2$, $y = 2x + 3$. What are the points of intersection?



a. $(3, 9)$ and $(1, 1)$

b. $(-1, 1)$ and $(1, 1)$

c. $(-3, 9)$ and $(1, 1)$

d. $(3, 9)$ and $(-1, 1)$

MCQ — Word problems: setting up and solving equations

88 A rectangular garden has a length that is 5 meters more than its width, and its area is 84 square meters. Find the width.

a. 12

b. 7

c. 9

d. 14

89 A mobile-data plan charges a fixed monthly fee of 40 riyals plus 2 riyals per GB used. If a user's bill was 68 riyals, how many GB did they use?

a. 34

b. 14

c. 54

d. 28